OpenGeoSys Training – KIGAM/ZJU 2026

Exercise 2: Fracture Flow - STIMTEC

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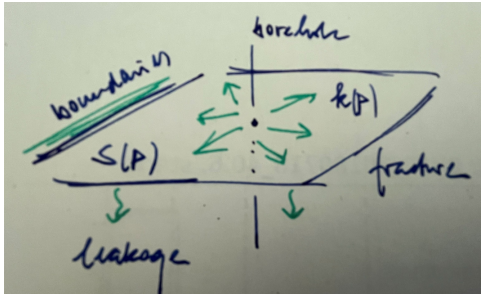
[4]Gesellschaft für Reaktorsicherheit – GRS, Braunschweig

Daejeon/Hangzhou, 14th October 2026

DECOVALEX 2027

Agenda

1 OGS Course Overview
2 References



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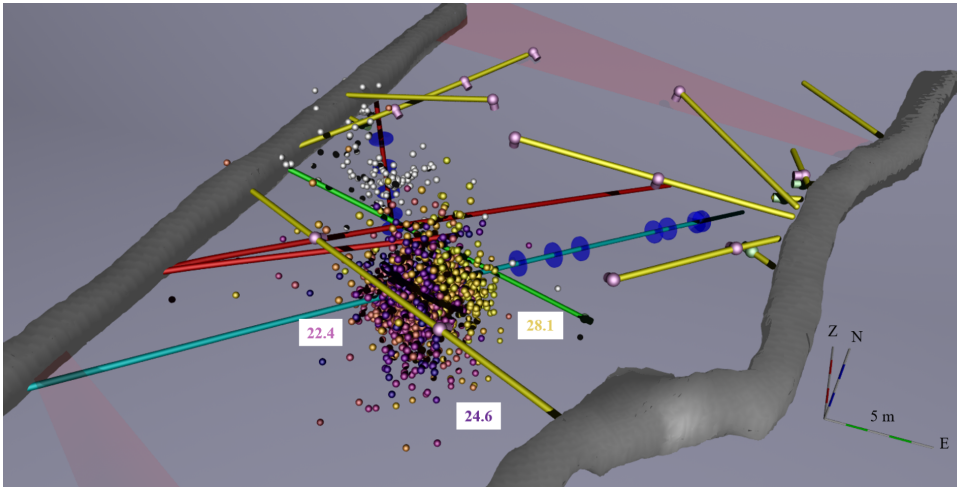
Exercise 3

Pumping tests

STIMTEC experiment

STIMTEC experiment

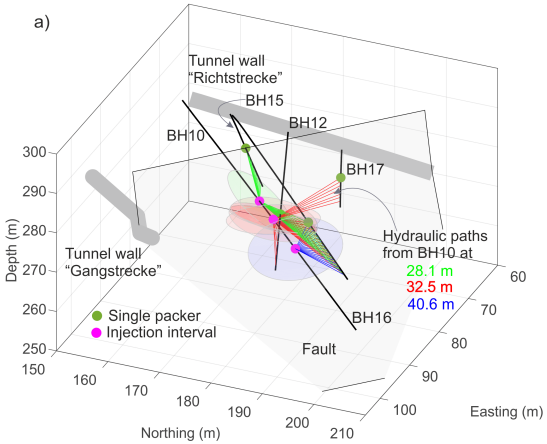
Underground Teaching and Research Laboratory "Reiche Zeche"



Sources: Renner et al. (RUB)

STIMTEC experiment

Fractured rock - pumping and monitoring wells



Sources: Renner et al.

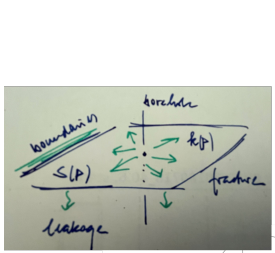
Hydraulic tests

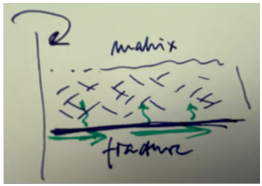
- various hydraulic behavior (permeability, storativity)
- BH10_20180716_51.6 (permeability)
- BH10_20180716_40.6 (storativity)

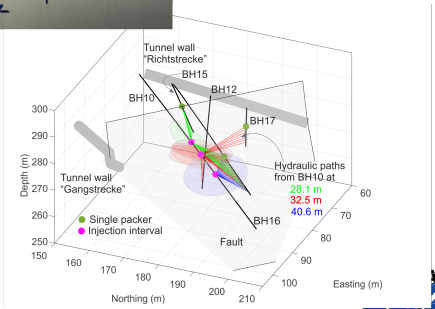
Modeling approaches:

- axial symmetry, fracture-matrix interaction
- single fracture (nonlinear fracture properties, leakage boundary conditions)
- discrete fracture network

We consider two pumping tests: BH10_20180716_51.6 and BH10_20180716_40.6, conducted at the same day but different borehole parts.

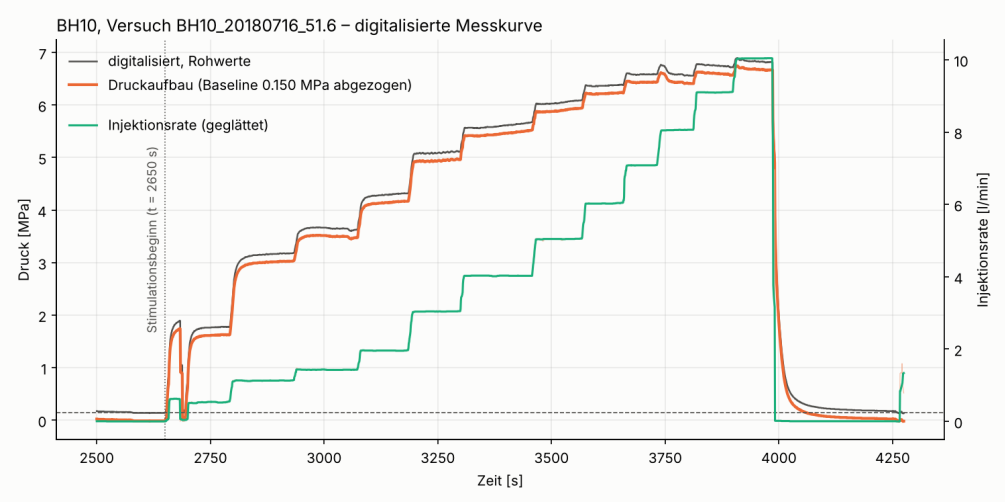






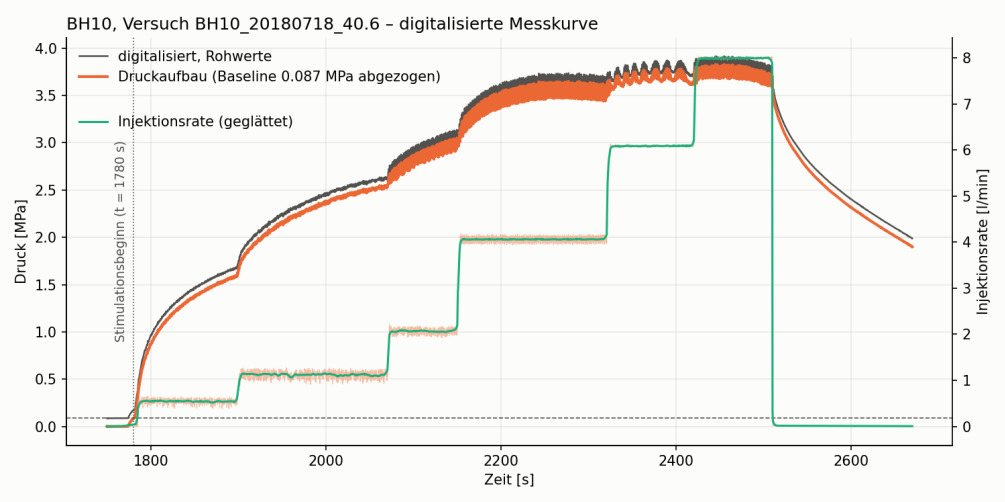
STIMTEC experiment: BH10_20180716_51.6

Hydraulic test



STIMTEC experiment: BH10_20180716_40.6

Modeling approaches: Hydraulic test



STIMTEC experiments

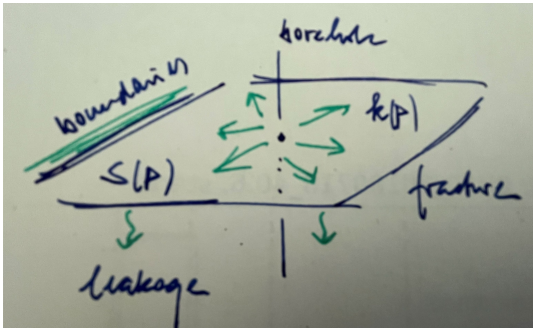
Exercise idea

- Step 1: ogs-stimtec-v1
  - run ogs project: single fracture with constant parameters k, S
  - given measured pressure increase data
  - ask AI to optimize parameters for best possible fit
  - discussion how to improve the model
- Step 2: ogs-stimtec-v2
  - ask AI to build pressure-dependent parameters k, S (with expressions)
  - ask AI to optimize parameters (first k, then both k and S)
- Step 3: post-processing
  - plot results at different observation points
  - plot contour lines and surfaces

GitHub repo:

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Model geometry



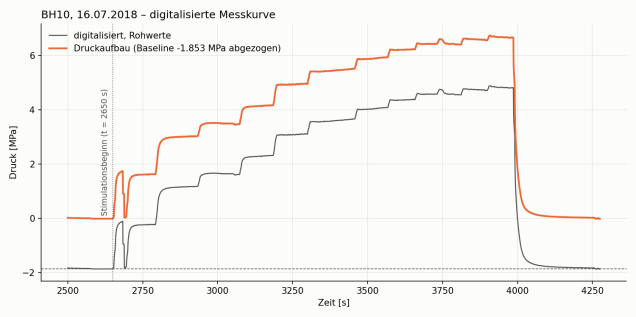
Model parameters

Parameter	Symbol	Value	Unit
Density	ρ	1000	kg m^3
Viscosity	μ	0.001	Pa s
Porosity	n	1	-
Permeability	k	expression	m^2
Storativity	S	expression	Pa^{-1}
Injection	q_{in}	expression	
Boundaries	BC	closed/open	
Leakage	q_{le}	function	

Table: Model parameters

Data preparation

Pumping tests: Pressure response



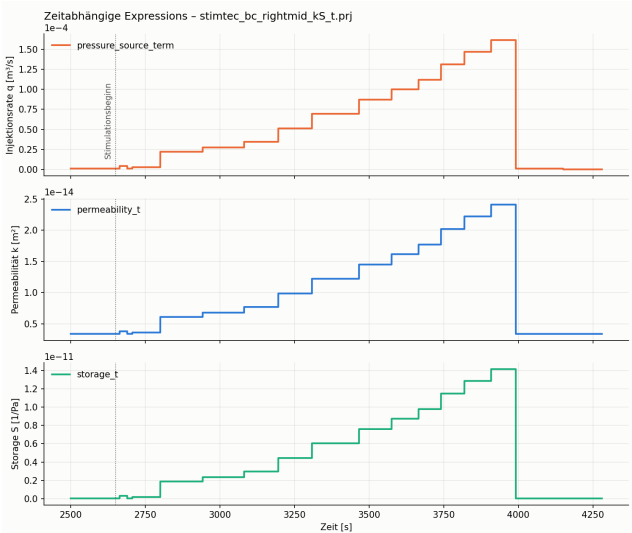
- Digitization of pumping curves: injection steps and measured pressure increase over time

Codes (subdir `workflow`)

→ `digitize_bh10.py`: Digitization of an original picture
→ `bh10_gemessen_digitalisiert.csv`: Excel file as result of digitization
→ `plot_messung.py`: Plot the excel files (from digitisation) properly

Data preparation

Pumping tests: Injection steps as `<expression>`



Fluid injection is represented by using **`<expressions>`** in the OGS project file.

Codes (subdir `workflow`)

```
<parameter>
    <name>pressure_source_term</name>
    <type>Function</type>
    <expression>
        (t &lt; 2663.923 ? 1.03227e-06 :
         t &lt; 2689.323 ? 4.20422763e-06 :
         ...
         t &lt; 4149.97 ? 9.76304533e-07 :
         0)
    </expression>
</parameter>
```

→ `plot_expressions.py`: Plot expressions for injections steps as well as related permeabilities and storages

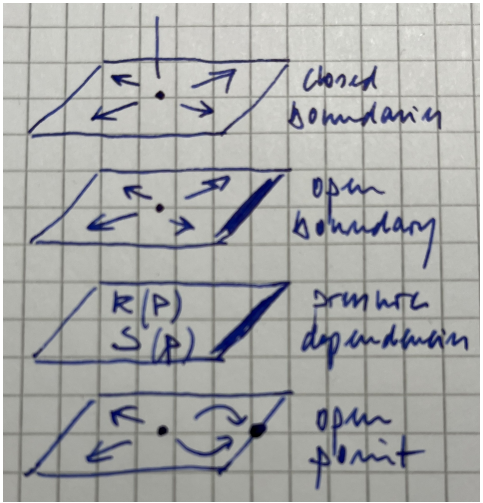
Exercise 3a

STIMTEC experiment

Fracture Model
(ToDo: Olaf)

Modeling Strategy

Pumping tests



1. Closed boundaries
   → `stimtec.prj`
2. Open boundary ($p = 0$ MPa) on the right side
   → `stimtec_bc_right.prj`
3. Pressure dependent permeabilities and storage coefficients according to the injections steps, and therefore, pressure increase
   → `stimtec_bc_right_kS_t.prj`
4. Limiting the open boundary part to a single point
   → `stimtec_bc_rightmid_kS_t.prj`

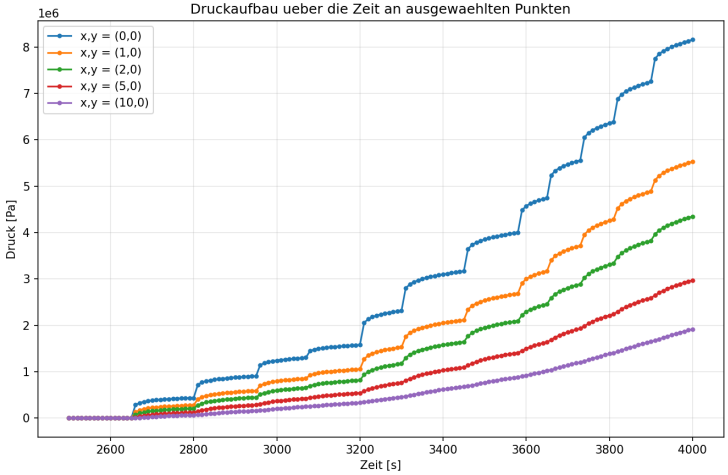
Codes (subdir `simulation`)



- Project: `prj`
- Geometry: `gml`
- Meshes: `vtu`

STIMTEC experiment: BH10_20180716_51.6

Modeling approaches: Single fracture



Constant permeability

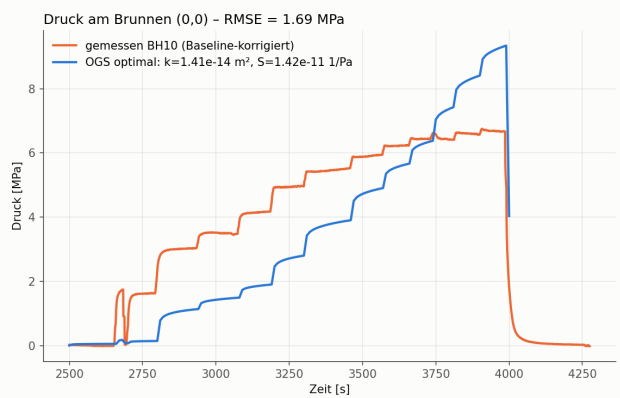
Pressure increase at different distances from the injection well

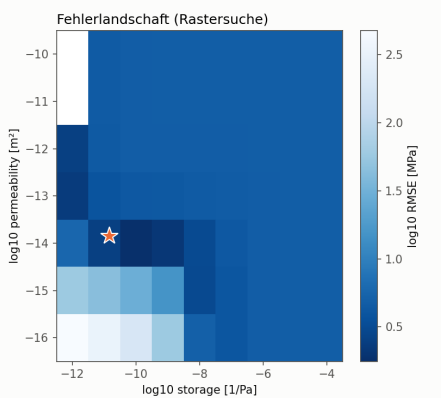
prj

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STIMTEC experiment: BH10_20180716_51.6

Modeling approaches: Single fracture, closed boundary



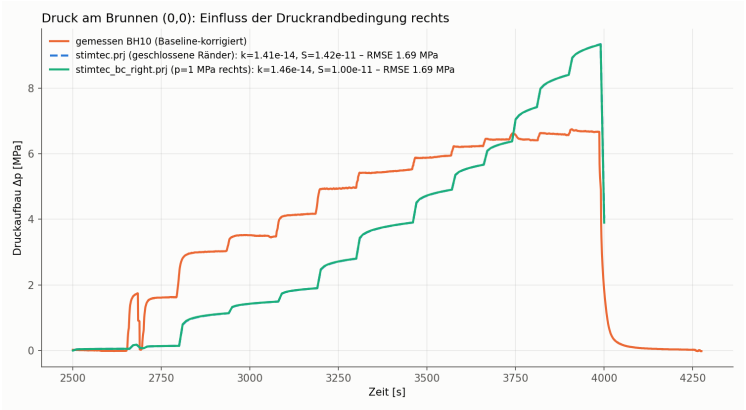


simulation/stimtec.prj

First calibration with constant permeability and storage (two-parameter optimization):
RSME = 1.69 MPa, $k = 1.409 \times 10^{-14}$ m^2, $S = 1.419 \times 10^{-11}$ 1/Pa

STIMTEC experiment: BH10_20180716_51.6

Modeling approaches: Single fracture, open boundary

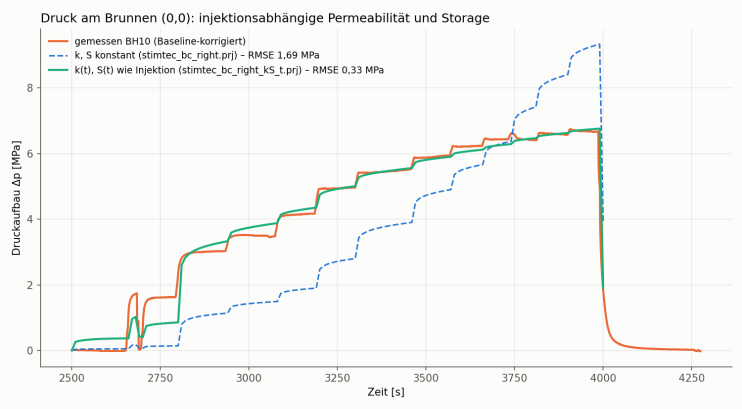


simulation/stimtec_bc_right.prj

Small effect of open boundary (only):
RSME = 1.685 MPa, $k = 1.460 \times 10^{-14}$ m^2 , $S = 1.000 \times 10^{-11}$ 1/Pa

STIMTEC experiment: BH10_20180716_51.6

Modeling approaches: Single fracture, injection-rate dependent permeability and storage



Decisive model improvement:

$k \in [3.3\times10^{-15}\text{-}2.61\times10^{-14}]$ in m^2

$S \in [3.02\times10^{-14}\text{-}5.22\times10^{-12}]$ in 1/Pa

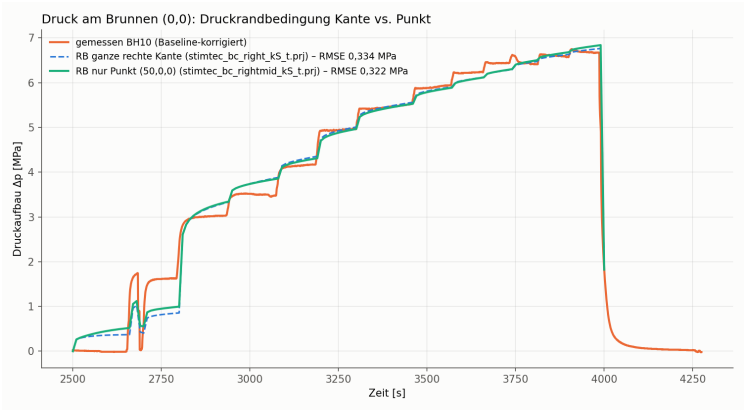
RSME = 0.33 MPa

Repo-Link

simulation/stimtec_bc_rightmid_kS_t.prj

STIMTEC experiment: BH10_20180716_51.6

Modeling approaches: Single fracture, injection-rate dependent permeability and storage, open point BC



Open boundary is reduced to a single point (very small outlet):

$k \in [3.37 \times 10^{-15}\text{-}2.41 \times 10^{-14}]$ in m^2

$S \in [2.48 \times 10^{-14}\text{-}1.41 \times 10^{-11}]$ in 1/Pa

RSME = 0.322 MPa

Repo-Link

`simulation/stimtec_bc_rightmid_kS_t.prj`

- Data field: `permeability_py`
- Data field: `storage_py`
- Mesh for data fields: `simulation/stimtec-square-py.vtu`
- Script: `workflow/ogs_py_kS.py`

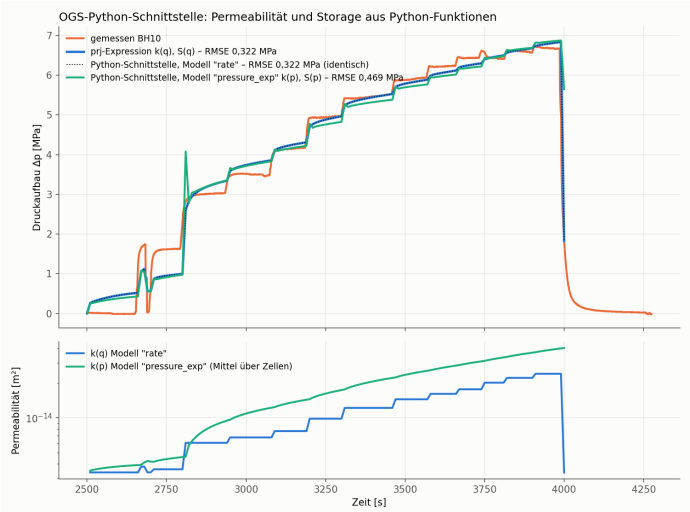
It performs calculations using the OGS Simulator, step by step. Before each step, Python calculates k and S and writes them directly to the OGS memory. It also reads the well pressure directly from the memory. One run takes approximately 1.5 s.

Execution: `py -3.13 ogs_py_kS.py`

`simulation/stimtec_bc_rightmid_py.prj`

STIMTEC experiment: BH10_20180716_51.6

Python interface



Latest version: Injection-rate dependent permeability and storage, open point BC

Two models

- `rate`: $k(q)$ and $S(q)$ exactly like expressions
- `pressure_exp`:

$$k = k_0 \exp \beta_k \Delta p$$

$$S = S_0 \exp \beta_S \Delta p$$

Repo-Link

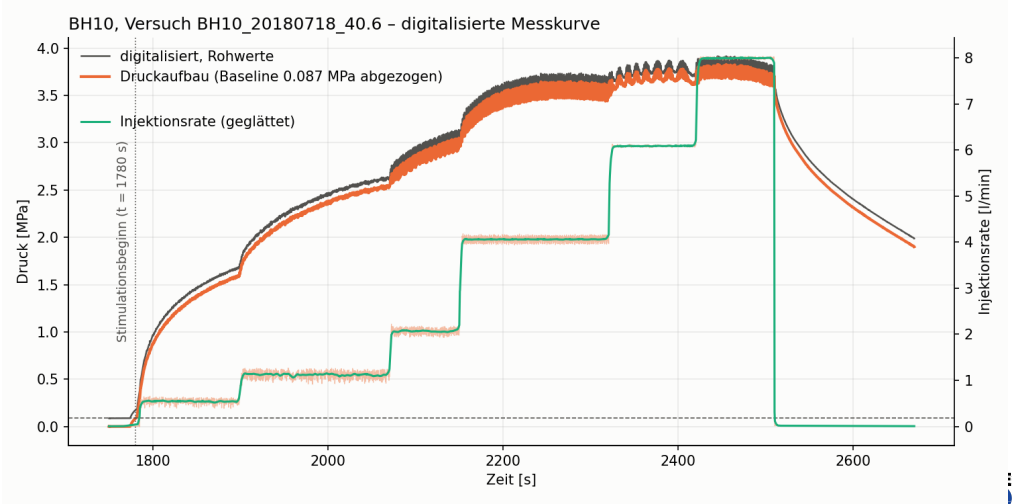
`py -3.13 ogs_py_kS.py`

Fazit:

- permeability increase with higher injection rates seems reasonable, as a (small) fracture opening can be expected with larger liquid pressure inside the fracture
- early stimulation period: $t \in$ [2654-2688] sec and [2689-2694] sec, needs to be considered explicitly (currently not differentiated)
- the two-parameter calibration seems better suited for STIMTEC experiment: BH10_20180716_40.6, i.e. smoother pressure build-up

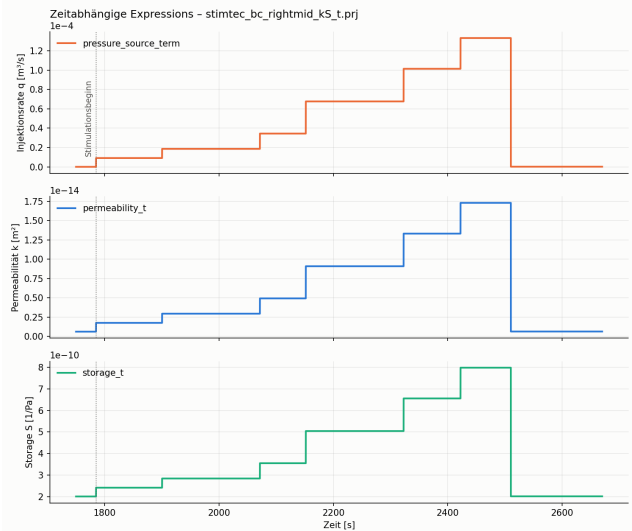
STIMTEC experiment: BH10_20180716_40.6

Hydraulic test



STIMTEC experiment: BH10_20180716_40.6

Data preparation: Injection steps as `<expression>`



Fluid injection is represented by using **`<expressions>`** in the OGS project file.

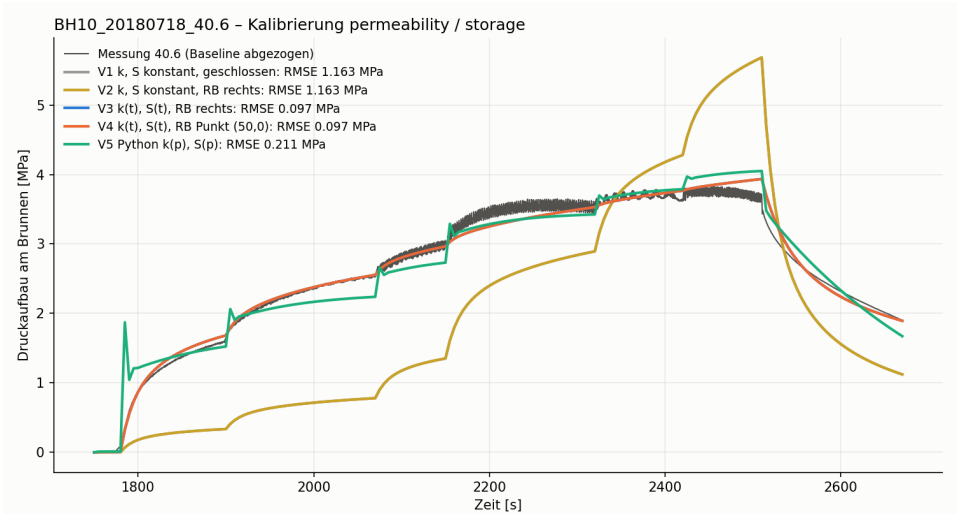
```
<parameter>
    <name>pressure_source_term</name>
    <type>Function</type>
    <expression>
        (t &lt; 2663.923 ? 1.03227e-06 :
         t &lt; 2689.323 ? 4.20422763e-06 :
         ...
         t &lt; 4149.97 ? 9.76304533e-07 :
         0)
    </expression>
</parameter>
```

Codes (subdir `workflow`)

→ `plot_expressions.py`: Plot expressions for injections steps as well as related permeabilities and storages

STIMTEC experiment: BH10_20180716_40.6

Modeling approaches: Single fracture



STIMTEC experiment: BH10_20180716_40.6

Modeling approaches: Single fracture: Best fit

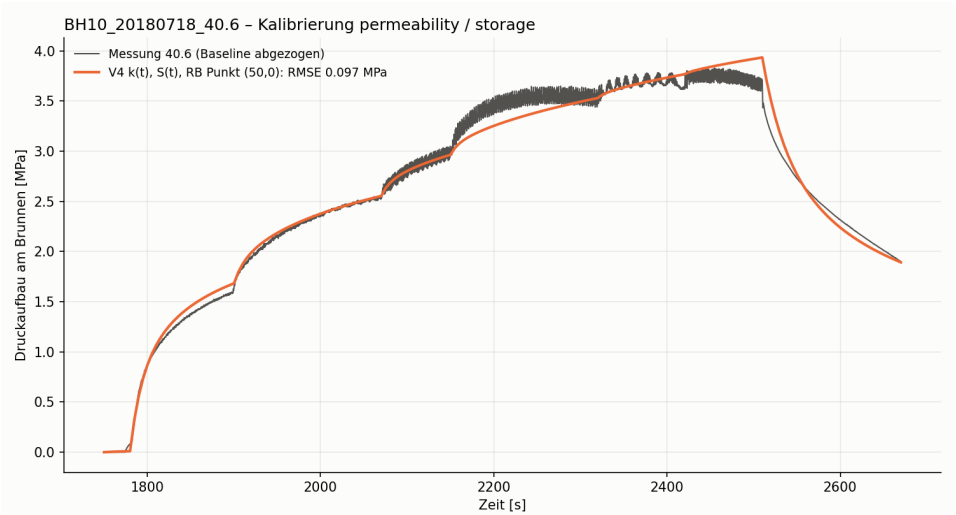


Table: Calibrated permeability k and storage coefficient S for injection test BH10_20180718_40.6 (OGS LiquidFlow 2D, 100 m × 100 m, point source at the well). Objective function: RMSE of the pressure build-up at the well, $t = 1750–2670$ s.

		k [m^2]		S [Pa^{-1}]		RMSE
Variant	BC[b]	k_{min}	k_{max}	S_{min}	S_{max}	[MPa]
V1: k, S constant	closed	9.15×10^{-15}		1.08×10^{-9}		1.163
V2: k, S constant	edge	9.15×10^{-15}		1.08×10^{-9}		1.163
V3: $k(t)$, $S(t)$	edge	6.16×10^{-16}	1.73×10^{-14}	2.01×10^{-10}	7.95×10^{-10}	0.097
V4: $k(t)$, $S(t)$	point	6.16×10^{-16}	1.73×10^{-14}	2.01×10^{-10}	7.98×10^{-10}	0.097
V5a: Python rate	point	6.16×10^{-16}	1.73×10^{-14}	2.01×10^{-10}	7.98×10^{-10}	0.097
V5b: Python pressure_exp[a]	point	$k_0 = 3.68 \times 10^{-15}$		$S_0 = 4.73 \times 10^{-14}$		0.211

V3–V5a: $X(t) = X_{min} + (X_{max} - X_{min})\,(q(t) - q_{min})\,/\,(q_{max} - q_{min})$ using the time intervals of the injection rate $q(t)$ (`pressure_source_term`); X_{min} after shut-in.

[a] $k = k_0 \exp(\beta_k \Delta p)$, $S = S_0 \exp(\beta_S \Delta p)$ per cell with $\beta_k = 1.06\,\text{MPa}^{-1}$, $\beta_S = 0.79\,\text{MPa}^{-1}$.

[b] Boundary condition: closed = no BC; edge = Dirichlet $p = 1$ MPa on the right edge ($x = 50$ m); point = Dirichlet $p = 1$ MPa only at point $(50, 0)$. Initial pressure equal to the boundary pressure.

Table: Comparison of injection tests 51.6 and 40.6 (V1: k, S constant; V4: $k(t)$, $S(t)$, point BC).

Quantity	Unit	51.6	40.6
Injection steps	–	15	6
Max. injection rate	l/min	10.0	8.0
Max. pressure build-up	MPa	6.8	3.8
V1: k	m^2	1.41×10^{-14}	9.15×10^{-15}
V1: S	Pa^{-1}	1.42×10^{-11}	1.08×10^{-9}
V1: RMSE	MPa	1.690	1.163
V4: $k_{\min}$	m^2	3.37×10^{-15}	6.16×10^{-16}
V4: $k_{\max}$	m^2	2.41×10^{-14}	1.73×10^{-14}
V4: $k_{\max}/k_{\min}$	–	7.2	28.1
V4: $S_{\min}$	Pa^{-1}	2.48×10^{-14}	2.01×10^{-10}
V4: $S_{\max}$	Pa^{-1}	1.41×10^{-11}	7.98×10^{-10}
V4: RMSE	MPa	0.322	0.097

Exercise 3b

STIMTEC experiment

Fracture-Matrix Model
(ToDo: Kristhian)

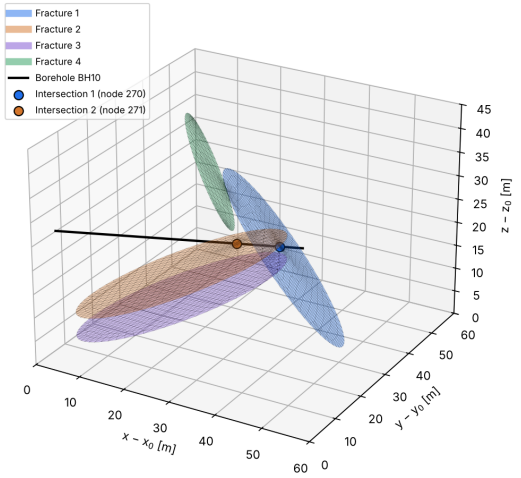
Exercise 3c

STIMTEC experiment

Fracture-Network Model
(ToDo: please review)

STIMTEC experiment: BH10_20180716_40.6

Modeling approach: Discrete Fracture Network (DFN) model



- DFN mesh `mesh.vtu` consists of 1834 nodes and 3440 linear triangles (mean edge length 1.8 m) representing four planar, approximately elliptical fractures embedded in 3D (MaterialIDs 1–4).
- Fracture 1 intersects all other fractures, whereas fractures 2, 3 and 4 do not intersect each other. Fractures 2 and 3 are parallel, gently dipping planes whose centres are offset vertically by 4.6 m; fractures 1 and 4 dip steeply in the same direction.
- Coordinates are given in Gauss–Krüger metres (origin of the plots: minimum coordinate of the mesh).

Table: Fractures of the DFN model (orientation from a plane fit through the nodes).

Fracture	Triangles	Nodes	Area [m^2]	Dip dir./dip [°]	Axes [m]	Intersected by BH10
1	1076	572	1599	053/53	53×38	yes (node 270)
2	1017	542	1434	270/24	44×41	yes (node 271)
3	882	475	1434	270/24	45×41	no (1.3 m off)
4	465	266	497	053/55	51×13	no ($>$ 10 m)

Flow in each fracture is described by the OpenGeoSys `LIQUID_FLOW` process on lower-dimensional (2D) elements in 3D space, in mass-balance form, with the fracture aperture a as a thickness factor:

$$a\,\rho\,S\,\frac{\partial p}{\partial t} - \nabla\cdot\left[a\,\rho\,\frac{k}{\mu}\left(\nabla p - \rho\,\mathbf{g}\right)\right] = \dot{m}, \tag{1}$$

with pressure p, specific storage S, permeability k, fluid density $\rho = 1000\,\mathrm{kg/m^3}$, viscosity $\mu = 1\times10^{-3}\,\mathrm{Pa\,s}$ and gravity $\mathbf{g} = (0, 0, -9.8)\,\mathrm{m/s^2}$. The injection enters as nodal mass sources $\dot{m}_i$ [kg/s] at the intersection nodes. Since both the storage and the conductance terms scale with a, the pressure response to a given injection rate scales with $1/a$; only the products $a\,k$ (transmissivity-type) and $a\,S$ (storativity-type) can be identified from the test.

With $\Delta p_1(t)$, $\Delta p_2(t)$ the simulated build-up at nodes 270 and 271 and $\Delta p_m(t)$ the measured build-up at the $N = 2852$ sample times:

$$\mathrm{RMSE}_{\mathrm{mean}} = \sqrt{\frac{1}{N}\sum_j \left(\tfrac{1}{2}\left[\Delta p_1 + \Delta p_2\right] - \Delta p_m\right)_j^2}, \tag{2}$$

$$\mathrm{RMSE}_{\mathrm{nodes}} = \sqrt{\frac{1}{2N}\sum_j \left[(\Delta p_1 - \Delta p_m)_j^2 + (\Delta p_2 - \Delta p_m)_j^2\right]}. \tag{3}$$

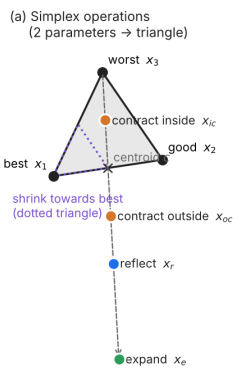
Both intersections lie within the same packed injection interval and must therefore experience the same pressure. $\mathrm{RMSE}_{\mathrm{nodes}}$ enforces this, whereas $\mathrm{RMSE}_{\mathrm{mean}}$ allows compensating deviations of the two intersections.

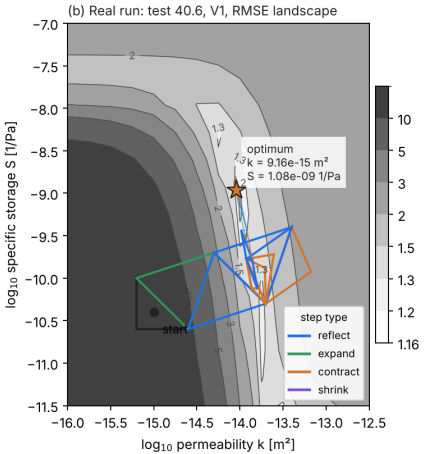
Nelder–Mead is a gradient-free method for minimising a function. It only needs function values, which suits our case: each evaluation is one complete OGS run, and we have no derivatives of the RMSE.
The idea: for n parameters it works with a simplex of n + 1 points. With our four parameters ($log\ k_{min}$, $log\ k_{max}$, $log\ S_{min}$, $log\ S_{max}$) that is 5 points; adding the rate split makes 6. Each step evaluates the objective at the points and moves the worst one:

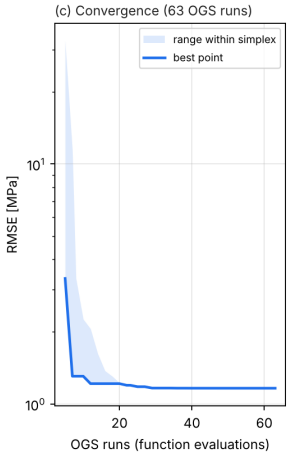
1. Order the points by objective value, from best to worst.
2. Reflect the worst point through the centroid of the others. If that point is better, the simplex has found a downhill direction.
3. Expand: if the reflected point is even better than the current best, go further in that direction.
4. Contract: if reflection gives no improvement, try a point between the worst point and the centroid.
5. Shrink: if contraction also fails, pull all points towards the best point.

STIMTEC experiment: BH10_20180716_40.6

Optimization procedure: Nelder-Mead





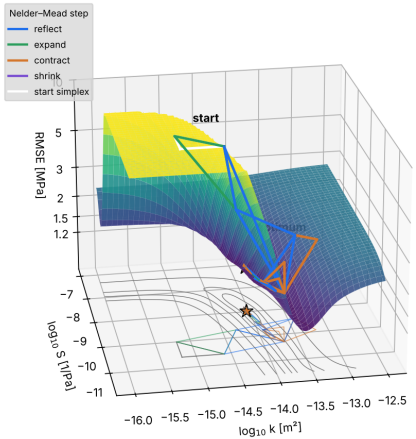


STIMTEC experiment: BH10_20180716_40.6

Optimization procedure: Nelder-Mead

RMSE landscape of test 40.6, variant V1 (constant k, S): 225 OGS runs; RMSE axis logarithmic, clipped at 10 MPa

(a) View along the valley (from low storage)



(b) View across the valley (from high permeability)

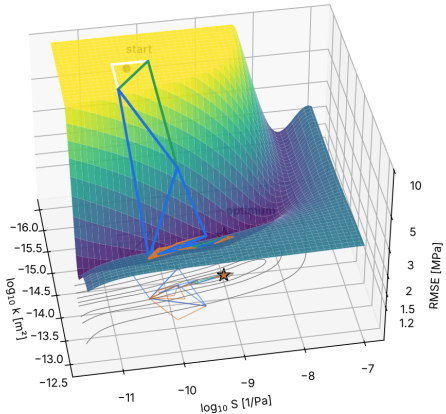


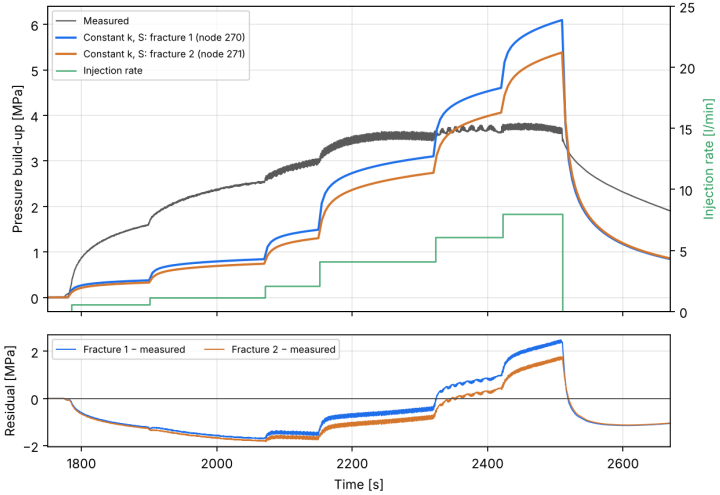
Table: Modeling strategy: Variants and 2D reference calibration of test 40.6.

Variant	Boundary	k [m^2]	S [Pa^{-1}]	RMSE [MPa]
V1: k, S constant	closed	9.15×10^{-15}	1.08×10^{-9}	1.163
V2: k, S constant	edge $p = 1$ MPa	9.15×10^{-15}	1.08×10^{-9}	1.163
V3: $k(t)$, $S(t)$	edge	6.16×10^{-16}–1.73×10^{-14}	2.01×10^{-10}–7.95×10^{-10}	0.097
V4: $k(t)$, $S(t)$	point $p = 1$ MPa	6.16×10^{-16}–1.73×10^{-14}	2.01×10^{-10}–7.98×10^{-10}	0.097
V5b: $k(p)$, $S(p)$[a]	point	$k_0 = 3.68 \times 10^{-15}$	$S_0 = 4.73 \times 10^{-14}$	0.211

[a] $k = k_0 \exp(\beta_k \Delta p)$, $S = S_0 \exp(\beta_S \Delta p)$ per element, $\beta_k = 1.06\,\text{MPa}^{-1}$, $\beta_S = 0.79\,\text{MPa}^{-1}$, explicit coupling via the OpenGeoSys Python interface.

STIMTEC experiment: BH10_20180716_40.6

Modeling strategy: V1

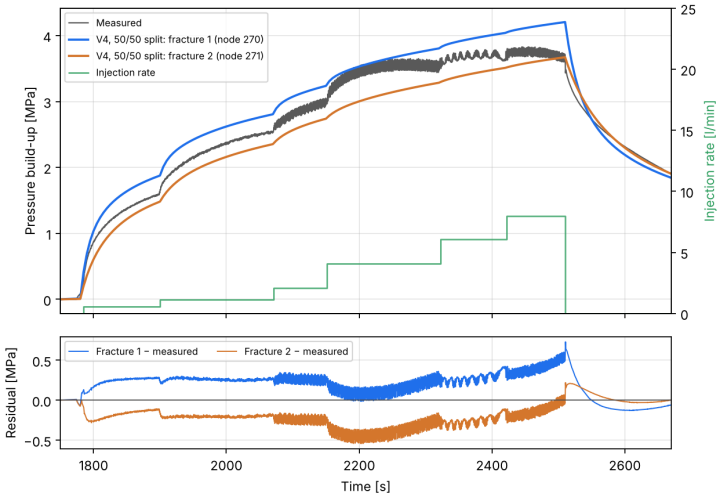


DFN model with constant k and S (means of 2D V4), $a = 1$ m, 50/50 rate split. $\mathrm{RMSE}_{\mathrm{mean}} = 1.24$ MPa.

Repo link

STIMTEC experiment: BH10_20180716_40.6

Modeling strategy: V4

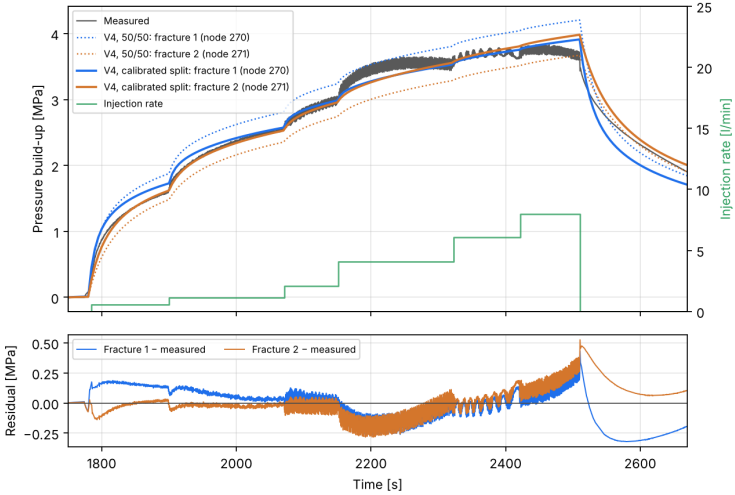


DFN model V4 ($k(t)$, $S(t)$ coupled to the injection rate), 50/50 rate split.
$RMSE_{mean}$ = 0.099 MPa,
$RMSE_{nodes}$ = 0.262 MPa.

Repo link

STIMTEC experiment: BH10_20180716_40.6

Modeling strategy: V4

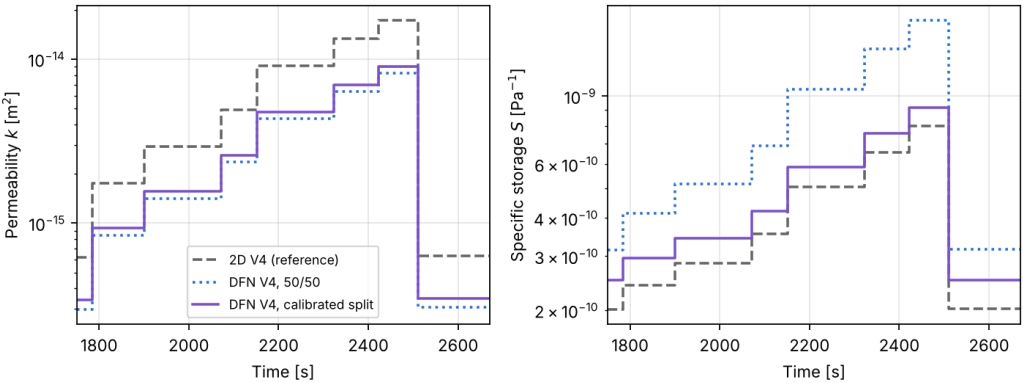


DFN model V4 with calibrated rate partitioning ($\alpha_1 = 0.465$; solid) compared with the 50/50 split (dotted). Residuals refer to the calibrated model. $RMSE_{nodes}$ = 0.118 MPa, $RMSE_{mean}$ = 0.102 MPa.

Repo link

STIMTEC experiment: BH10_20180716_40.6

Modeling strategy: V4



Calibrated permeability and specific storage as functions of time (step-wise) for the 2D reference V4 and the two DFN variants (values for unit thickness / $a = 1$ m).

STIMTEC experiment: BH10_20180716_40.6

OGS-DFN Workflows

File	Content
`find_bh10_intersections.py`	Line–triangle intersection of BH10 with the DFN
`BH10_intersections.vtu/.csv`, `BH10_frac1.vtu`, `BH10_frac2.vtu`	Point sub-meshes at nodes 270/271
`bc_rim.vtu`	262 rim nodes of the fractures (Dirichlet p_0)
`make_prj_dfn_st.py`	Nodal source terms instead of Neumann on BH10 (`-rim`: rim BC)
`make_prj_dfn_40.py`	Injection steps 40.6, constant k, S, aperture option
`make_prj_dfn_40_kS_t.py`	V4 parameterisation $k(t)$, $S(t)$; options `-split1`, `-calib`
`calibrate_dfn_kS_t.py`	Calibration of k, S ($\mathrm{RMSE}_{\mathrm{mean}}$)
`calibrate_dfn_kS_split.py`	Calibration of k, S, α_1 ($\mathrm{RMSE}_{\mathrm{nodes}}$)
`*_log.csv`, `*_best.txt`	Optimisation logs and optima
`stimtec_dfn_40_a1.prj`	Constant k, S, $a = 1\,\mathrm{m}$
`stimtec_dfn_40_kS_t.prj`	V4, 50/50 split
`stimtec_dfn_40_kS_split.prj`	V4 with calibrated split (best variant)

Typical calls (in `stimtec/dfn`, with ogs 6.5.9 on the path):

```
python make_prj_dfn_st.py stimtec_dfn.prj stimtec_dfn_st_rim.prj --rim
python make_prj_dfn_40.py stimtec_dfn_st_rim.prj stimtec_dfn_40_a1.prj \
       --aperture=1
python calibrate_dfn_kS_t.py
X0="-15.5265,-14.0834,-9.5033,-8.7521,0.47" OBJ=nodes \
       python calibrate_dfn_kS_split.py
python make_prj_dfn_40_kS_t.py stimtec_dfn_40_a1.prj \
       stimtec_dfn_40_kS_split.prj --kmin=3.3989e-16 \
       --kmax=9.0585e-15 --smin=2.5043e-10 --smax=9.1470e-10 \
       --split1=0.4652
ogs stimtec_dfn_40_kS_split.prj -o out
```

Repo link

References

[1] Olaf Kolditz. Computational Methods in Environmental Fluid Mechanics. Springer Berlin Heidelberg, 2002. isbn: 9783662047613. doi: 10.1007/978-3-662-04761-3. url: http://dx.doi.org/10.1007/978-3-662-04761-3.

[2] Erik Nixdorf, Mike Sips and Peter Morstein. 'Development of a digital workflow for layer reduction strategies and visual-analytical outcome analysis to enhance geo-hydraulic modelling efficiency'. In: Digital Water 3.1 (2025), pp. 1–24. doi: 10.1080/28375807.2025.2460825.

[3] Charles V. Theis. 'The Relation Between the Lowering of the Piezometric Surface and the Rate and Duration of Discharge of a Well Using Ground-Water Storage'. In: Transactions, American Geophysical Union 16.2 (1935), pp. 519–524. doi: 10.1029/TR016i002p00519.

[4] Zhi-Yuan Zhang, Christian Schmidt, Erik Nixdorf, Xingxing Kuang and Jan H. Fleckenstein. 'Effects of Heterogeneous Stream-Groundwater Exchange on the Source Composition of Stream Discharge and Solute Load'. In: Water Resources Research 57.8 (Aug. 2021), e2020WR029079. doi: 10.1029/2020WR029079.